

PROJECT SYNOPSIS

Video Sharing Platform

A Web-Based Application for Uploading, Sharing and Watching Videos Online

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1. Introduction

A Video Sharing Platform is a web-based application that allows users to upload, share, and watch videos online. Just like popular platforms such as YouTube, this project aims to build a simple system where users can create accounts, upload their own videos, and watch videos uploaded by other users. The platform will also allow users to like, comment, and share videos with others.

2. Objective

- To design a user-friendly platform for uploading and watching videos.
- To allow users to create accounts and manage their profiles.
- To provide features like search, like, comment, and share.
- To store and manage video data efficiently using a database.
- To ensure secure login and data privacy for all users.

3. Scope of the Project

- Provides a simple and user-friendly platform for sharing videos online.
- Allows students, content creators, and small communities to upload and access video content.
- Supports core video-sharing features such as video upload, streaming, likes, comments, and subscriptions.
- Offers secure user authentication and video management functionality.
- Focuses on essential features without the complexity of large-scale video-sharing platforms.

Future Scope:

- Integration of live streaming functionality.
- Development of a mobile application version.
- Implementation of AI-based video recommendation systems.
- Addition of monetization features for content creators.

4. System Requirements

4.1 Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB or more
- Hard Disk: 50 GB free space (for storing videos and database)
- Internet Connection: Required for hosting and accessing the platform

4.2 Software Requirements

- Operating System: Windows 10 / Linux
- Frontend: HTML, CSS, JavaScript, React
- Backend: Node.js .
- Database: MongoDB / MySQL
- Tools: Visual Studio Code, Git, Postman

5. Proposed Modules

5.1 User Module

- User registration, login, and profile management.

5.2 Video Upload Module

- Upload, edit, delete, and categorize videos.

5.3 Video Playback Module

- Stream videos and view related content.

5.4 Interaction Module

- Like, comment, and share videos.

5.5 Search Module

- Search videos by title, category, or uploader.

5.6 Admin Module

- Manage users, moderate content, and view platform statistics.

6. Working of the System

The system works in the following simple steps:

1. A user registers or logs into the platform.
2. The user can upload a video by adding a title, description, and category.
3. The video is stored on the server and linked to the user's account in the database.
4. Other users can browse, search, and watch the uploaded videos.
5. Viewers can like, comment, or share the videos they watch.
6. The admin monitors the platform to ensure content follows guidelines.

7. Advantages of the Proposed System

- Easy and simple interface for all types of users.
- Provides a platform for new content creators to showcase their work.
- Helps users learn full-stack web development concepts.
- Can be customized and extended with new features in the future.
- Cost-effective compared to large commercial platforms.

8. Conclusion

The Video Sharing Platform project is developed to provide an easy and efficient way for users to upload, share, and watch videos online. It helps users interact through features like likes, comments, and subscriptions while offering a simple and user-friendly experience. Through this project, we gain hands-on experience in web development, database management, and software design. The platform serves as a strong foundation for future improvements, such as live streaming, mobile application support, and AI-based video recommendations.